

Roberto Paolo Jiménez Moreno

A01251931@tec.mx | +52 6624300568 |

<https://www.linkedin.com/in/roberto-paolo-jim%C3%A9nez-moreno-1b4330212>

Education

Tecnológico de Monterrey (ITESM)

Mechatronics Engineering student. Cumulative score 98.93

Hermosillo, Son

June 2024

Relevant Coursework:

- Development of research in optimization of Solar Panel controllers in collaboration with Professeur Driss Mehdi and Dr, Luis Carlos Felix Herran, based on Neural Network algorithms from 2022-2023. **Development of testing scenarios and passed the investigation to new students for collaboration.**
- Project in TE AI Cup to generate a welding classifier with convoluted neural networks.
- Prototype of cyber physical system of a box palletizer for TE Connectivity with VR immersive experience included. **Winner of “ExpolIngenierías”**
- Development of a row machine with air resistance, and interface for counting calories and exercise time. **Participation in “ExpolIngenierías”**
- Development of web page proposals for companies “FairPlay” and “The Garden ”, a flower shop.

Social Commitment projects:

Collaboration with non-profit institutions such as: “Ganfer”, “AGP Asociación George Papanicolau”, “CEFORMA”, “Grameen de la frontera” and even creating my own social service “Ruta del Emprendimiento” based on the creation of entrepreneurship courses for teach them at public schools, we were able to give some entrepreneur lessons to different COBACH Highschools.

Professional Experience

Instituto de Emprendimiento Eugenio Garza Lagüera

Software Engineer & Project Specialist

Hermosillo, Sonora

01/2021 - 12/2022

- Regional BTEM (Beca al talento emprendedor) project coordination, automation of physical processes, verification and proposal of an improvement program. Creation, maintenance and updating of the Change Makers Day Campus Sonora Norte 2021 page and creation of a page to visualize the progress of different entrepreneurs. Generation of holograms and acrylic projections. Digital coordination of the Select USA event with the US embassy.

GSE Biomed

Prototype Engineer Intern

Hermosillo, Sonora

08/2022 - 12/2022

- Responsible for electronic prototyping, programming and 3D modeling. Programming in Python, C, and C++.

MAGNA SFT

Control Engineer Intern

Hermosillo, Sonora

06/2021-07/2021

- PLC's assembly and programming, Verification and correction of electronic and programming errors, Building of control circuits. Basic knowledge of interconnection of manufacturing areas.

TE CONNECTIVITY

Manufacturing Engineer Intern

Hermosillo, Sonora

01/2023 - 05/2023

- Generation of continuous improvement projects, Automation of processes in production cells 3D design and process optimization.

TESLA INC.

Controls Engineer Intern

Fremont, California

08/2023 - 02/2024

- Basic Control Automation, Drivers replacement and configuration, Grafana Dashboard creation, MES system implementation, PLC programming (Allen Bradley, Siemens and Beckhoff), sensor installation and creating and modifying HMIs.

Entrepreneurship Experience

Co-Founder of +Sangre (2020-2021)

An entrepreneurship focused on automating the blood donation process, aiming to reward altruists and create a more supportive community towards blood donation.

Notable achievement: Being among the finalists in the INNOVA Sonora competition in 2020.

Founder of Free Motion (2021-2022)

An entrepreneurship that creates, through artificial vision and mechatronic coupling, a mobility device for people with almost any type of paralysis or degenerative disease.

Notable achievement: Representing the Sonora Norte campus in the national Intra Hub competition by Enactus, in the highest category of entrepreneurship. The event was held in Aguascalientes in 2022.

Co-Founder of Airgraph (2021-2022)

An entrepreneurship aiming to create a device for all households that acts as a graphene-based filter, ionizing to achieve 99% elimination of viruses and bacteria.

Notable achievement: Being one of the highlighted projects at the ÍMPETU event of the leadership area.

Co-Founder of Enhanced Surgical (2024)

An entrepreneurship that develops mixed reality systems for the training, planning, and execution of surgeries, through the use of augmented reality headsets.

Notable achievement: Being selected as representatives for the Hult Prize in Guadalajara and advancing to the second stage in Boston (June 9, 2024), as well as establishing a connection with the deans of medicine at Tec Salud campus Guadalajara, having a community of 20 specialized surgical doctors, and a focus group of approximately 100 medical students for validations.

Leadership & Activities

ATHENA Leadership Representative Team

Leader of projects based on Sustainable Development Goals and Technological Innovation. Generation of cross-cutting social impact projects to support the community. During my tenure on the leadership representative team, I was always recognized for my consistent compliance with team standards and readiness, successfully participating in projects such as the previously described Airgraph project, in which I was the leader, and a prior project focused on generating energy from organic waste biomass (ECOMASS), one of the first projects of the degree and organizer of the team. I also collaborated in the Blue Fest event.

CORE Student Group

A student group focused on the development of young entrepreneurs on campus, directly linked with the Eugenio Garza Laguerre Entrepreneurship Institute. This group primarily focuses on disseminating information, training through individual sessions, and networking for young entrepreneurs, to later catalyze and train them to be incubated by the entrepreneurship institute and follow up on these initiatives. I became Vice President of this group in 2021 and later President in 2022.

ENCORE Student Group

Eventually, we sought to expand the student group and strengthen it with external organization, leading to a merger between the CORE and Enactus groups to form this group. While it may not seem like a major change, as these were primarily based on entrepreneurship, the main difference is that we aimed to generate enterprises with a social impact but a sustainable business model, with the main goal of training new entrepreneurs with a business foundation to create entrepreneurial leaders with social impact.

Skills & Interests

Certifications & Evidences::

https://drive.google.com/drive/folders/1-8xyTSwhBUGMSWS269jtjaFFe_W158-6?usp=sharing

Technical: Programming knowledge in Python, SQL, HTML, CSS, JS, C# and C++.

Process automation, whether digital or mechanical.

Allen Bradley , Siemens, Beckhoff and Delta PLC programming.

Use of prototyping machinery such as laser cutting, 3D printing, and CNC machines.

HMI Development, Network Configuration and Electronic Control systems.

Artificial Vision / Artificial intelligence (Neural Networks) / Machine Learning Models

Cyber Physical VR, Simulations and environment creation (Unity)

Management of SQL databases.

Data Analysis.

Solid Works design.

Soft skills:

Leadership and coordination of projects

Adaptability, versatility and problem resolution

Ability to implement ideas or proposals for innovation around technology.

Collaboration, teamwork and planification

Self-taught and auto-analysis

Commitment, responsibility and discipline

Language: Spanish (Native) | English C1 (TOEFL Certification) | French B1 (DELF Certification) | German B1 (Level 6 at Unison, pendant certification)

Interests: Project development, software learning, GYM (exercising), Self Development and Leadership.